Apex3D

Checkpoint 2 - Transformation and Animation

This evidence pack demonstrates how the group applied transformations and animation in Pure C++20 to simplified Formula 1 components developed during Checkpoint 1. All images were generated by the repository's CPU-rendered demonstration programs.

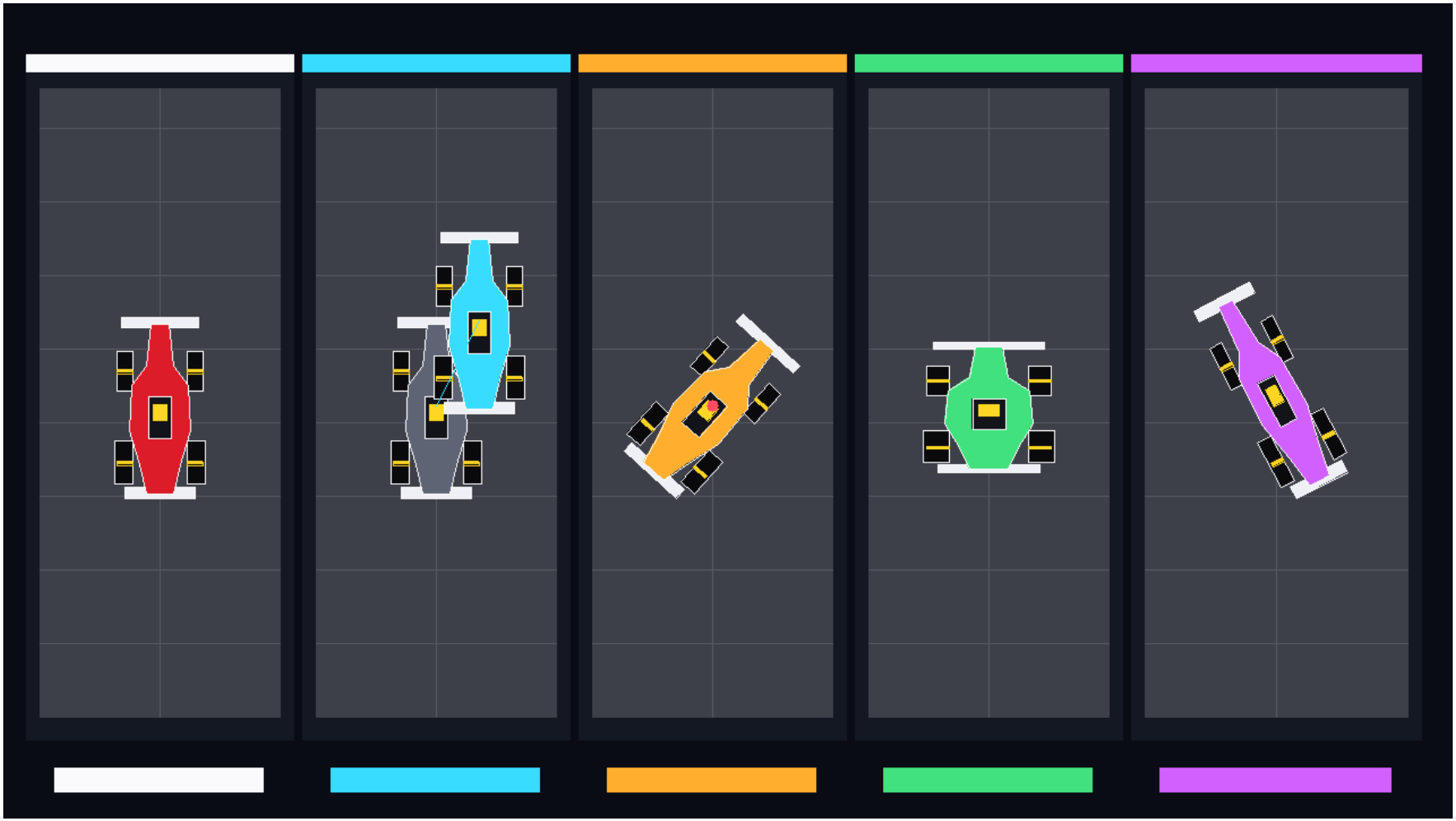
Implemented techniques

- 3 x 3 homogeneous matrices and composed 2D transformations
- Frame timing, keyframes and position/angle interpolation
- Hierarchical wheel rotation and rolling without slipping
- Translation, rotation and scaling of a multi-polygon car
- Catmull-Rom spline motion with tangent-based orientation
- Cubic Bezier motion with approximate arc-length sampling

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Verification: checkpoint2/verify_all.sh - all strict builds, tests and demonstrations pass

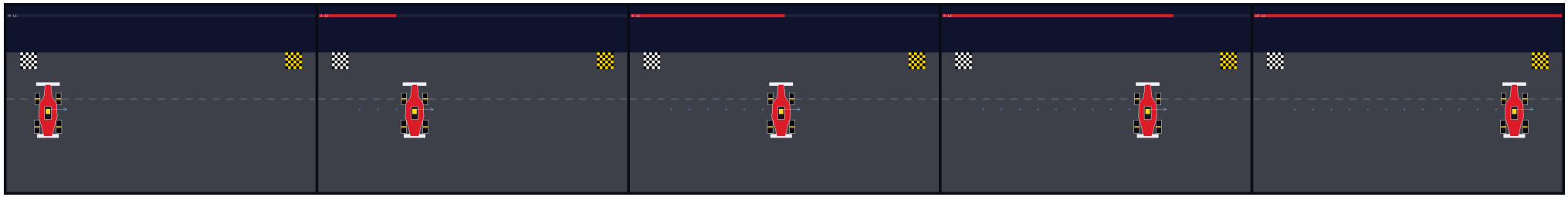
1. Homogeneous 2D transformations



A simplified Formula 1 car is transformed using handwritten 3 x 3 homogeneous matrices. The demonstration compares the original model with translation, rotation, scaling and a composed transform.

8 deterministic transformation tests pass.

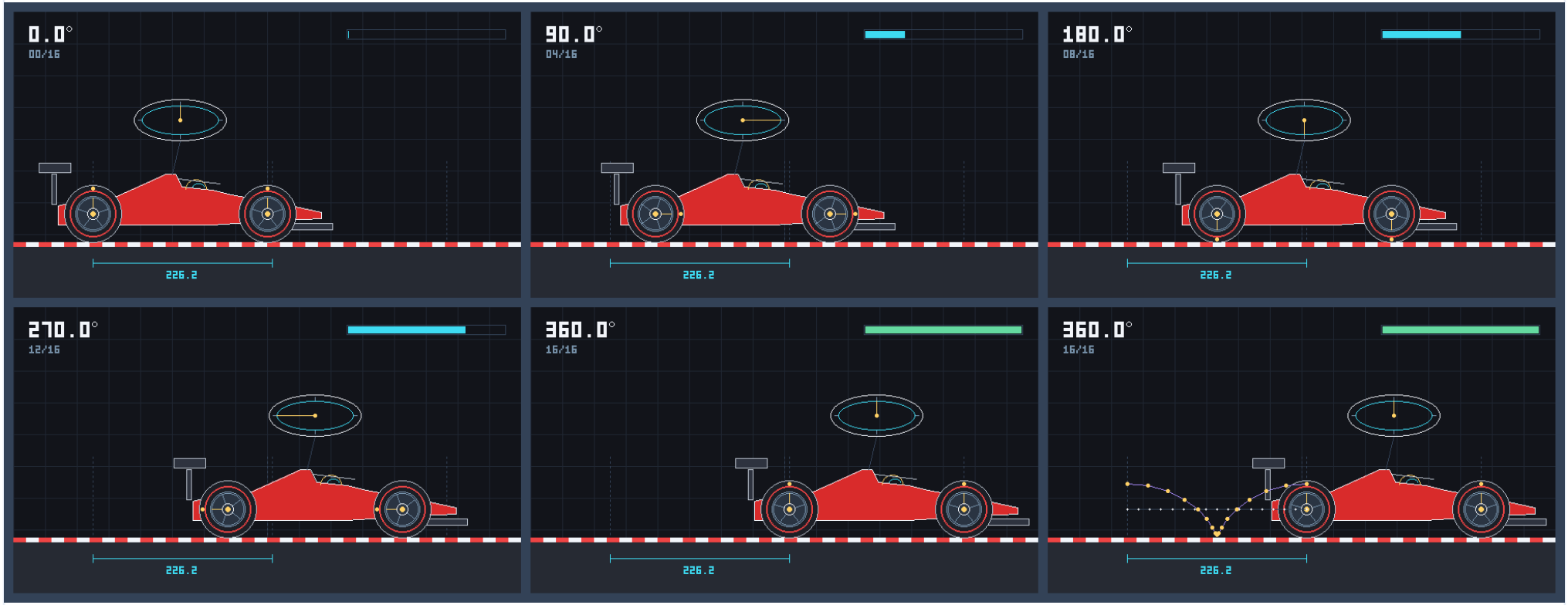
2. Timeline and interpolation



A frame clock converts frame numbers into animation progress. Linear position interpolation moves the project component across 13 frames, with clamping, looping and shortest-arc angle interpolation tested.

23 deterministic timeline and interpolation checks pass.

3. Hierarchical wheel animation



Circle and ellipse components from Checkpoint 1 are animated as wheels and a telemetry pod. Parent translation and local wheel rotation are combined, including rolling distance equal to radius times angle.

17 hierarchy, geometry and rendering checks pass.

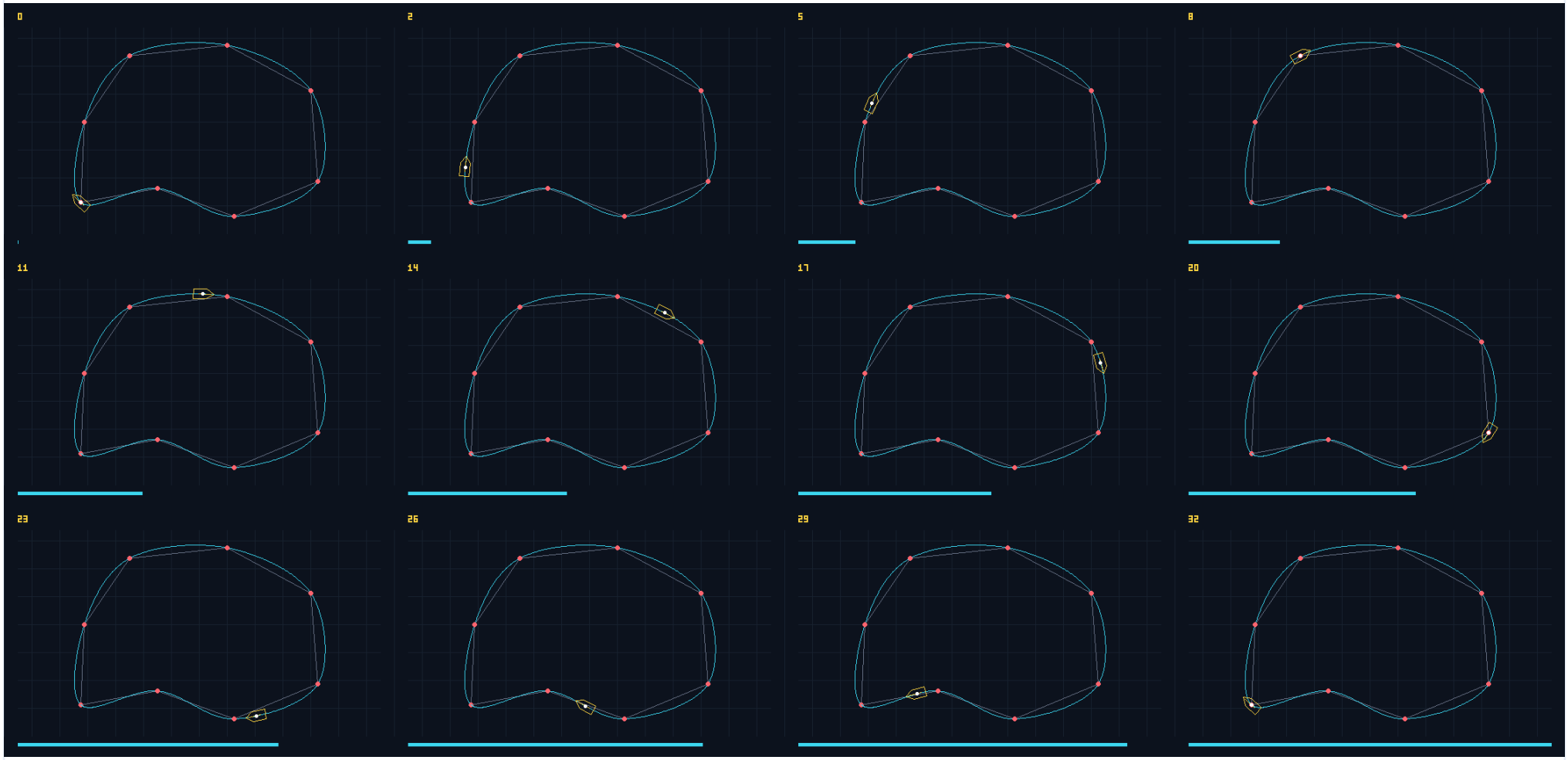
4. Keyframed polygon car



Thirteen polygons form a simplified Formula 1 car. One shared matrix animates the complete model through translation, centre-pivot rotation, non-uniform scaling and combined motion across 25 frames.

11 pose, geometry, raster and PNG checks pass.

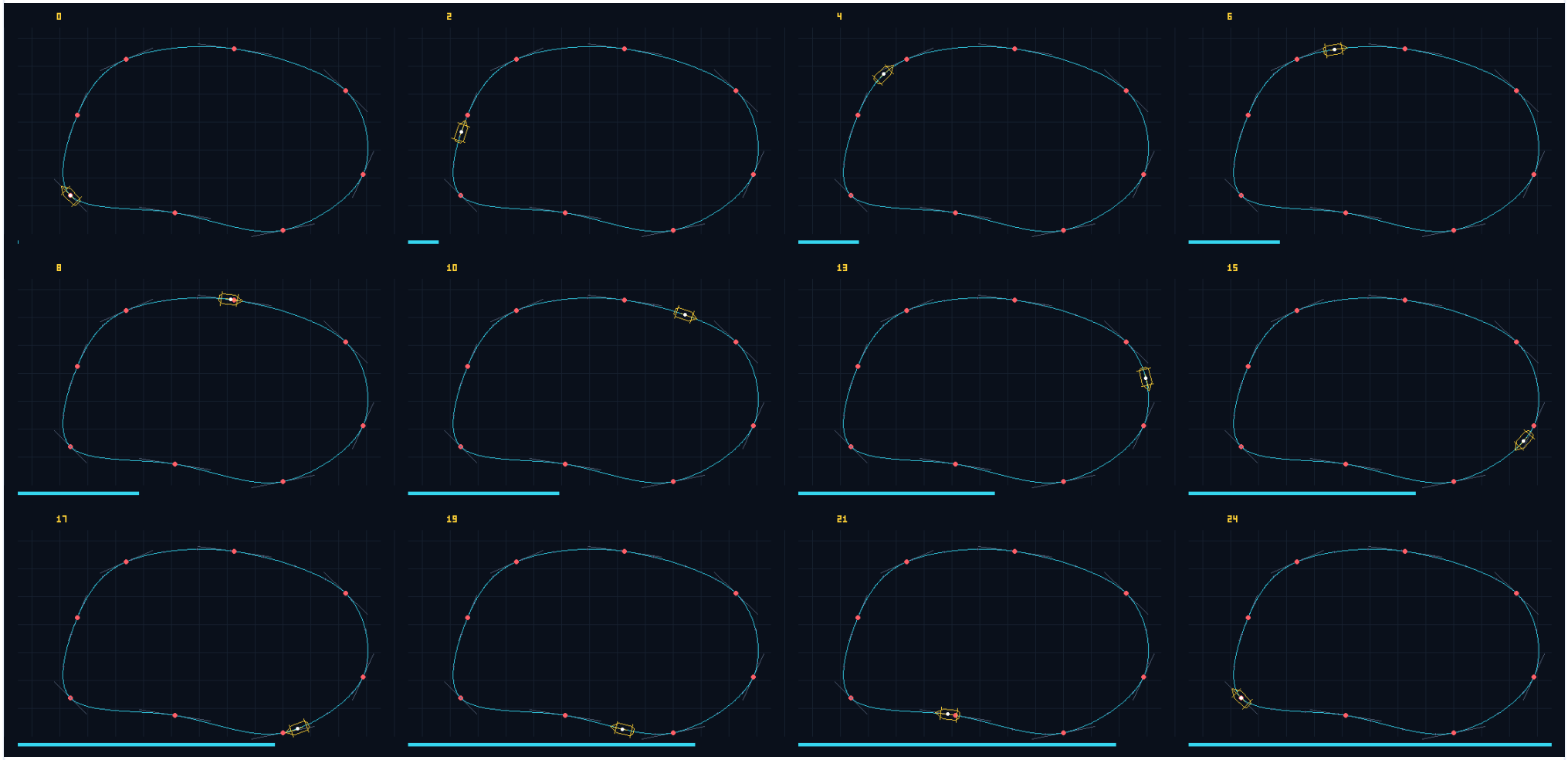
5. Catmull-Rom spline motion



A closed Catmull-Rom spline represents a simplified circuit path. An analytic tangent determines the car heading so the vehicle follows and faces along the curve throughout a 33-frame lap.

All spline motion checks pass for 8 segments and 33 poses.

6. Cubic Bezier path animation



Eight C1-continuous cubic Bezier segments define a closed circuit. An analytic derivative controls heading, while an arc-length lookup table keeps the car's 25-frame travel approximately uniform.

14 curve, continuity, motion, pose and rendering checks pass.